

VISA INC. INTERVIEW EXPERIENCE 2022

BATCH: 2020 – 2024

SELECTION PROCESS FOR: Summer Internship

NAME OF THE COMPANY: VISA Inc., Bangalore

ABOUT THE COMPANY:

Visa Inc. is an American multinational financial services corporation headquartered in San Francisco, California. It facilitates electronic funds transfer throughout the world, most commonly through Visa-branded credit cards, debit cards and prepaid cards. Visa is often misinterpreted as a credit card company, but it is not so. Visa is the world's leader in digital payments. Its mission is to connect the world through the most innovative, reliable and secure payment network – enabling individuals, businesses, economies to thrive.

NUMBER OF ROUNDS: 2

1. Online Assessment – Coding round
2. Interview – Technical + HR

ONLINE ASSESSMENT – CODING ROUND:

Platform: HackerRank

Duration: 60 minutes

Number of questions: 2 Coding questions (no aptitude questions)

Level of difficulty: Easy – Medium

The test was held on 22nd of August, 2022. It was conducted in online mode under the invigilation of college officials, where the students were provided systems at computer lab of IST Department to attend the test. Everyone got different set of questions and the following were the 2 questions that I got:

- Given two arrays named a and b of size n and m respectively. Find the maximum number of pairs (a[i], b[j]) such that $a[i] > b[j]$, $0 \leq i < n$ and $0 \leq j < m$. Also, an element of the array can appear at-most in only one pair.
 - Example 1: Array a = {1, 2, 3, 4, 5}; Array b = {1, 6, 1, 6, 1};
Answer: Maximum Possible Pairs = 3
Explanation: The possible pairs are: (2, 1), (3, 1), (4, 1)
Note: Many such sets of possible pairs can be found. But the question here is to find the number of such possible pairs and not to find the pairs.
 - Example 2: Array a = {3, 2, 3}; Array b = {8, 1, 2};
Answer: Maximum Possible Pairs = 2
Explanation: The possible pairs are: (2, 1), (3, 2)

This was an easy question, and my solution passed all the testcases given.

- One unit of an alloy can be made from n different metals. The composition of the n different metals required for alloy preparation will be given in composition matrix of size

n. The current stock availability of each metal and the cost of buying one unit of a metal will be given in stock matrix and cost matrix, each of size n . Given a budget (≥ 1), the task is to find the maximum number of units of alloy that can be prepared by utilizing the available stock and buying more stock within the budget.

○ Example 1:

Value of $n = 3$; Budget = 14;

Composition = {2, 1, 2}; Stock = {1, 0, 0}; Cost = {2, 2, 1};

Answer: Maximum number of units that can be produced = 2

Explanation: 3 different metals A, B, C are required for alloy preparation. One unit of alloy can be prepared by 2 units of Metal A + 1 unit of Metal B + 2 units of Metal C. Currently, the available stock is: 1 unit of Metal A, 0 units of Metals B and C. Now, we can buy 1 unit of Metal B + 2 units of Metal C for a total cost of 6 rupees ($1*2 + 2*2$) and make the first unit of alloy. We are still left with a budget of 8 rupees. Now we can buy, 2 units of Metal A + 1 unit of Metal B + 2 units of Metal C for a total cost of 8 rupees and produce the second unit of alloy. We cannot buy more metals; thus, we stop here.

○ Example 2:

Value of $n = 4$; Budget = 45;

Composition = {3, 2, 3, 1}; Stock = {2, 6, 5, 4}; Cost = {4, 2, 1, 3};

Answer: Maximum number of units that can be produced = 3

Explanation: Cost required for producing first unit of alloy = 4

Cost required for unit 2 production = 13

Cost required for unit 3 production = 15

Cost required for unit 4 production = 22

Thus, only 3 units can be produced ($4+13+15 \leq 45$).

This was a medium level question, and my solution passed 10/15 testcases given. The results were declared on 25th August 2022. A total of 14 students were selected for the next round.

INTERVIEW:

The interview was conducted on MS Teams, and was about 45 minutes long. It was held on 26th August, 2022. Initially the interviewer asked me if I had any question to ask him before we moved on to the actual interview.

He then asked me about the projects that I did till now in college. I briefly explained each of my three projects that I had put on my resume. Then, he asked me what are the subjects that I have studied till now. I listed few courses (OS, DBMS, CA). He then asked me which one of those was my favourite subject. I said I liked the Operating Systems course much, and gave him a satisfactory reason for choosing it. He then asked me about the topics that I studied in OS. I listed a few topics (general introduction to OS and its functionalities, threads, process synchronization, deadlocks and virtual memory). The follow back questions on the topics were:

- What is a thread? Difference between a process and a thread.
- What are deadlocks?
- What is the difference between mutex and semaphore?

With this, we moved on to the coding pair of the interview. The HackerRank link was shared for the code-pair part. There were totally 4 questions in this part of the interview.

- Consider the below pseudocode and predict the output if main function is called.

```
public class abc {  
    public void f() {  
        f();  
    }  
    public void main() {  
        f();  
    }  
};
```

We had a long discussion on this question, wherein I was not able to come up with the correct answer the first time. But, with the hints given by the interviewer, I was able to answer it up to some point, after which we moved on to the second question.

- The second question was "Detect loop in a linked list" - <https://www.geeksforgeeks.org/detect-loop-in-a-linked-list/>. The interviewer explained me the question well with 2 examples, and asked me to code. I explained a simple approach, which will work if the maximum limit of nodes is known and then explained the fast and slow pointer method to detect a loop in the linked list. (Always start with the naïve approach and try to optimize it further). The interviewer told to just write the function code/pseudocode for the function which takes in the head pointer of linked list as argument and returns 1, when there is a loop found, else returns 0. I coded the function by explaining each line while I was typing. The interviewer seemed satisfied on my solution. He also asked about the time and space complexity of my solution.

- What will be the output of the following cases?

i. int a = 10; int b = 20;
...
if(a==11 && b==20) { ... }
else { ... }

Which part of the if-else statement will be executed and what are all the expressions that are evaluated (whether b==20 will be executed or not)?

ii. int a = 10; int b = 20;
...
if(a=11 && b==20) { ... }
else { ... }

What will be output now and what happens here?

iii. int a = 10; int b = 20;
...
if(a=11 & b==20) { ... }
else { ... }

What will be the final evaluation of condition given inside the if statement?

All these questions were on the basics of C Programming and I answered these questions satisfactorily.

- What will be printed if this program gets executed?

```
void main() {  
    int* b = f();  
    printf("%d", *b);  
}  
int* f() {  
    int c = 10;  
    return &c;  
}
```

For this question, I was not able to come up with the correct answer the first time. But the interviewer gave me hints to come up with the correct answer. After that, I was able to provide a correct answer, with which the interviewer seemed satisfied.

With this the interview got over. Before leaving, he asked me if I had any questions to ask. I asked him two questions which I prepared beforehand. He answered my questions and took leave. The results were out by 10 pm the same night. And, finally 11 of us were selected for the summer internship programme.

TOPICS TO FOCUS:

- Data Structures and Algorithms
- Operating Systems
- Database Management Systems
- Object Oriented Programming Concepts
- Whatever you've mentioned in your resume

RESOURCES THAT WERE USEFUL FOR PREPARATION:

- "Must to do coding questions" available in GeeksforGeeks - <https://www.geeksforgeeks.org/must-do-coding-questions-for-companies-like-amazon-microsoft-adobe/>
- Resources shared by ACM – Student Chapter of CEG through CODE Intern Preparation programme.

KEY TAKEAWAYS:

- Always be in touch with DSA. Keep solving coding questions on platforms like LeetCode, HackerRank, GeeksforGeeks and HackerEarth. Consistency in solving coding questions is an important factor. Do try to solve 2 to 3 problems everyday if possible.
- Know to code from scratch as well. Some companies expect you to code from scratch in the online assessment. So, kindly make sure you are thorough with the header files and syntax of the programming language well. (If you are using STL in cpp, make sure you know the header files for each of the elements used in the program and the syntax for each of the functions used with the template).
- Once you have been shortlisted for the interview round by any company, kindly go through the interview experiences of that particular company available online. "GeeksforGeeks Archives (or) Interview Experiences" helped me a lot to get a basic idea

on how the interview will proceed. Not only for the interviews, this helps for the online assessment rounds as well.

- Prepare for “Tell me about yourself” beforehand. Make sure it is short and crisp.
- Read and re-read your resume. If questioned on your resume (projects as well), you should be able to answer it satisfactorily. So, be thorough with whatever you’ve mentioned in your resume.
- Before attending any interview, do research on the company’s profile. This gives you a basic idea on how the company functions, it’s mission and goal as well. If the interviewer asks “What do you know about the company”, you should be able to answer it.
- Be honest and don’t panic. Stay calm even if you don’t know an answer for a question. If struck at any point, don’t hesitate to say “I’m not sure about this, I will definitely look into it”. The interviewer will help by giving hints as well.
- Don’t lose hope during the interview, thinking that you have not done well so far. This may affect you throughout the rest of the interview and that is not what the interviewer expects from you.
- During the code-pair round, if you are asked to code, explain the lines of code while you are coding. This gives the interviewer an idea that you are going with the flow of the approach and you have understood the approach well. Also, give sensible names for identifiers and functions during code-pair part of the interview.
- Prepare well for “HR Questions” too. Along with the general HR questions, based on the soft skills mentioned on your resume, you might be expected to narrate an incident where you think you possessed the particular skill. So, be prepared with narrating such incidents as well.
- Always stay motivated and optimistic throughout the period of preparation. Never get upset, if you don’t get through the process of any company. You will definitely get a better company than that.

✨ ALL THE BEST ✨

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